CptS 317. 10/14.

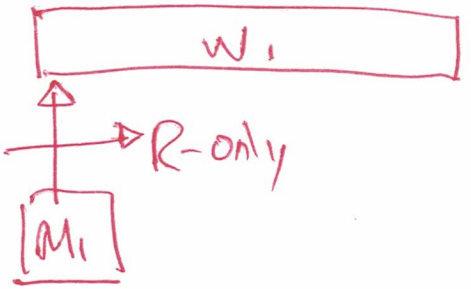
Automata-To-Automata Construction.

(hard problem). Let L_1 and L_2 be two reg languages. Define

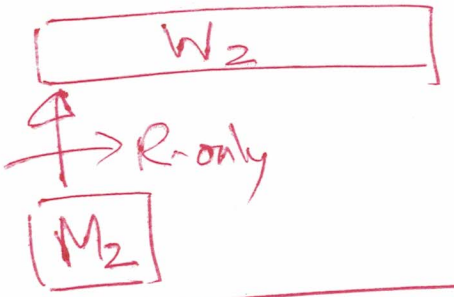
$$L_1 \bowtie L_2 = \{ w : \exists w_1 \in L_1 \text{ and } \exists w_2 \in L_2 \text{ s.t. } w \text{ is a suffix of } w_1 \text{ and } w \text{ is a prefix of } w_2 \}$$

Show that $L_1 \bowtie L_2$ is also reg.

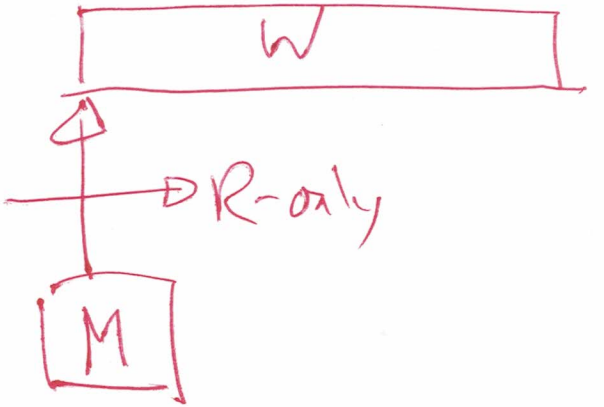
M_1



M_2



M



$\exists\, w_1, w_2$:
$w_1 \in L_1$ and $w_2 \in L_2$ and
w is a suffix of w_1
w is a prefix of w_2

$\Downarrow$

iff

M says yes on w.

$\exists w_1, w_2$: (says yes on)

M_1 accepts w_1 and

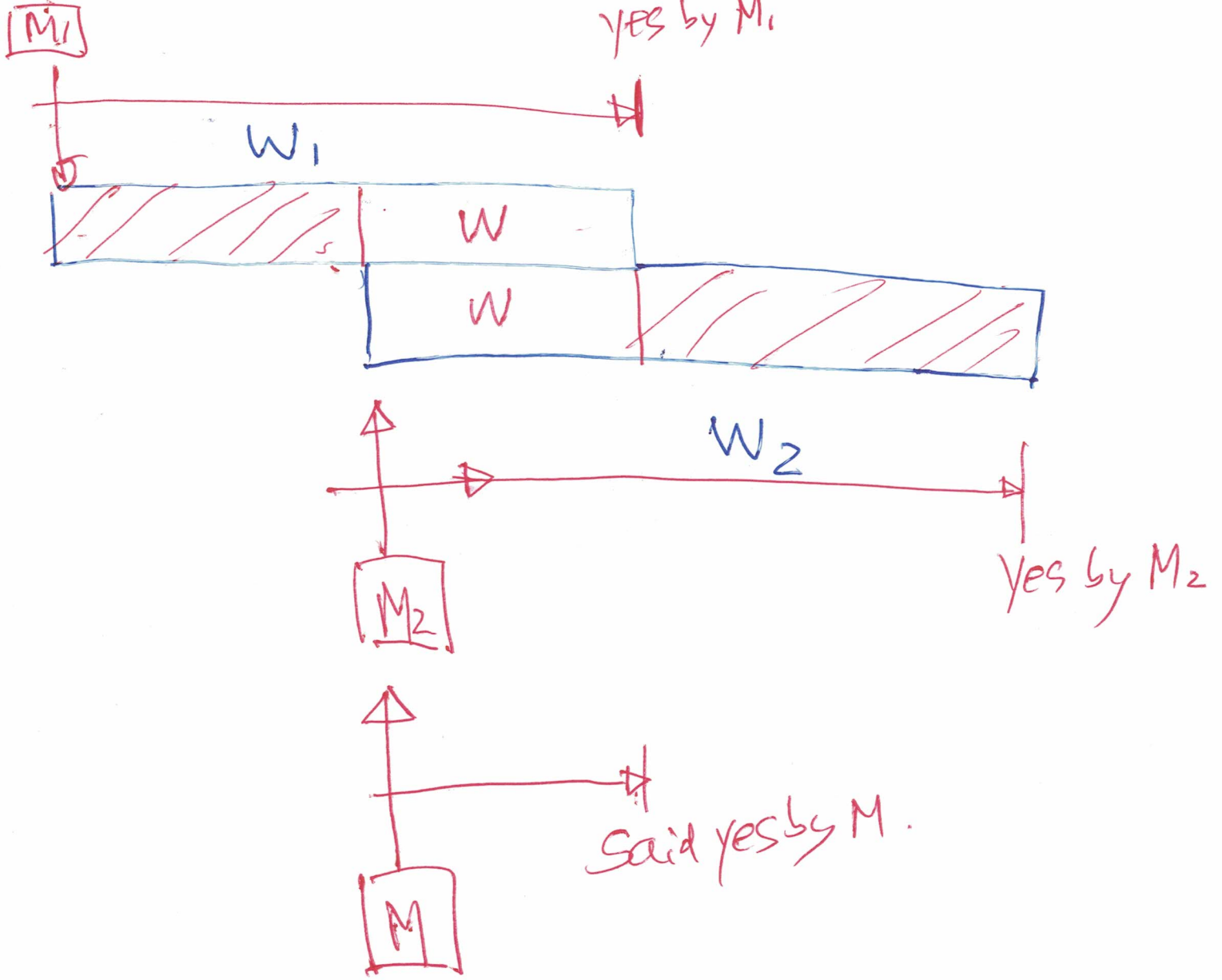
M_2 accepts w_2 and

w is a suffix of w_1

and w is a prefix of w_2

iff M says yes on w.

Proof. Since L_1 and L_2 are reg, we have FA M_1 and M_2 to accept them, resp. Now, we construct a FA M to accept $L_1 \cup L_2$. On input w, M runs as follows. Before M reads w, M guesses a symbol at a time and lets M_1 run on the guessed symbols. Some moments later, M nondet. stops the guess and starts to read the w. From this time, M simulates both M_1 and M_2 running on the w, in parrallel while M reads the w. At the end of w, M guesses a symbol at a time and lets M_2 continue to run on the guessed symbols. Some


M1
yes by M1
W1
W
W
W2
M2
yes by M2
M
said yes by M.

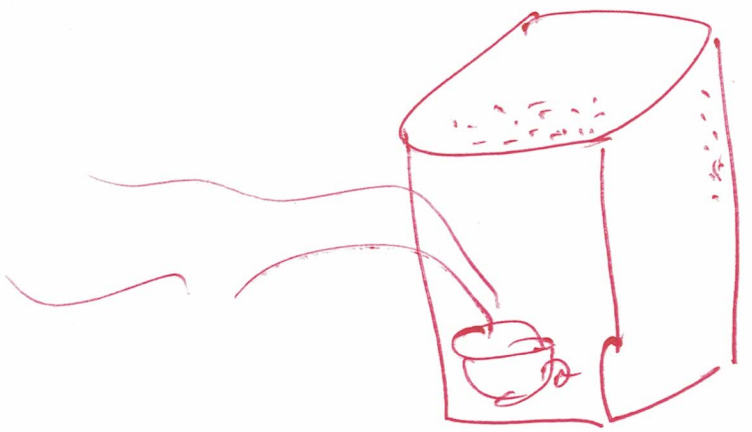
moments later, M guesses now M_2 shall stop. At that moment, M says yes on w if both M_1 and M_2 say yes. Clearly, M accepts $L_1 \cup L_2$. $\blacksquare$

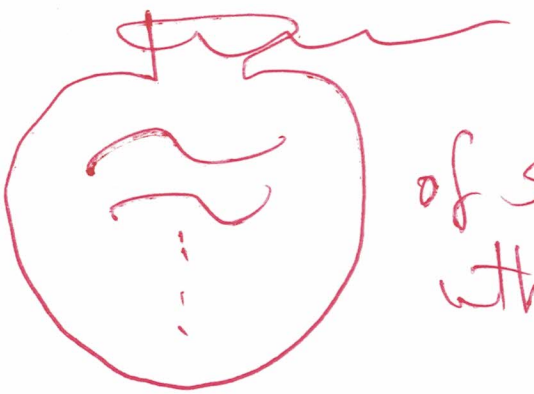
Summary: FA, RegExp, Pumping lemma.

Emptiness of FA
Infiniteness of FA } have algs to solve.

Use it! Example.

Given: a coffee maker. Play with it for 10 yrs.





of strings with len 10.

I have a pattern P on strings.

Set-up: Given: a Bag of strings
a pattern P.

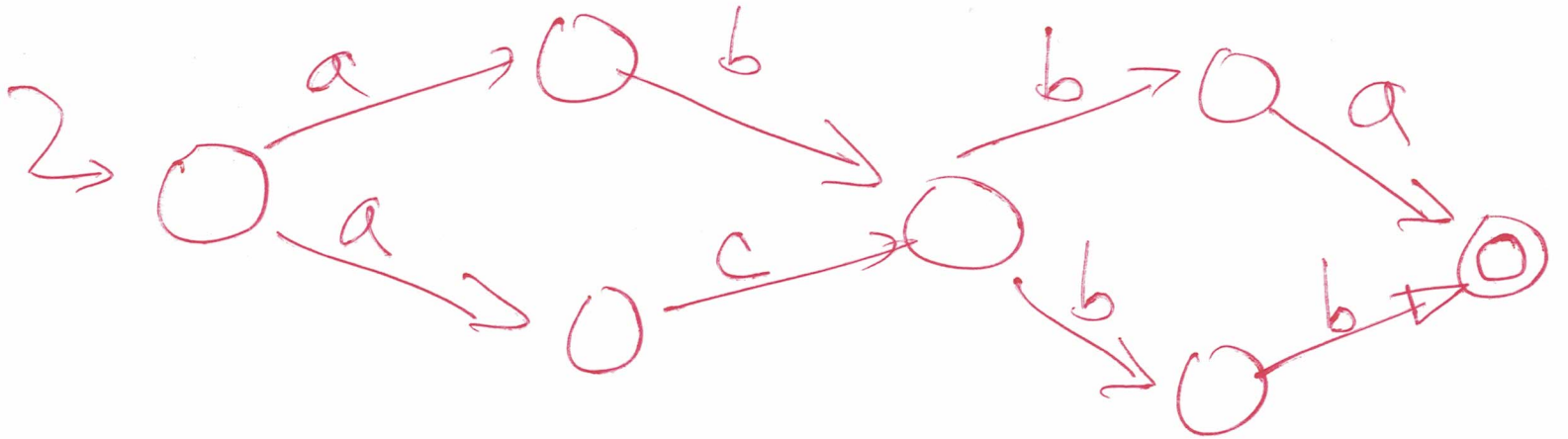
Question: Is there a string ~~that do~~
in the Bag that doesn't sat.
the pattern P ?

Alg: Construct a FA M_{Bag} for the Bag;
Construct a FA $M_{\neg P}$ for the pattern $\neg P$;
Construct FA $M = M_{Bag} \times M_{\neg P}$.

Cartesian Product

Check emptiness of M.

Example:



Bag of strings:

abba,
abbb,
acba,
acbb

FA is a data structure for
a set of sequences.